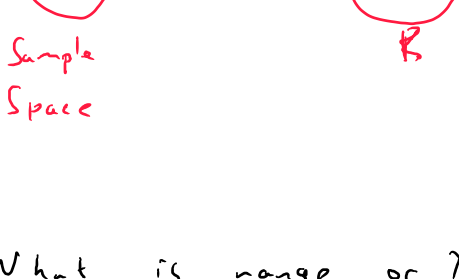


Random variables

- Random variable is a real valued function that assigns numerical values to possible outcomes of the random experiment.

A random variable X is a function from sample space to real numbers

$$X: S \rightarrow \mathbb{R}$$



Ex: Toss a coin 3 times

X is the number of heads. What is range of X

$$\text{Range}(X) = R_X = \{0, 1, 2, 3\}$$

H H H $\rightarrow 3$

H H T $\rightarrow 2$

H T T $\rightarrow 1$

T T T $\rightarrow 0$

T H H $\rightarrow 2$

T T H $\rightarrow 1$

T H T $\rightarrow 1$

H T H $\rightarrow 2$

Ex: Toss a coin 100 times, X is the number of tails

$$R_X = \{0, 1, 2, \dots, 100\} \rightarrow \text{Countable} \rightarrow \text{finite}$$

Ex: I toss a coin until the first tail. Let Y be the number of tosses. $R_Y = \{1, 2, \dots, \infty\} \rightarrow \text{Countable} \rightarrow \text{infinite}$

$\Rightarrow$ if the range R is countable, X is a discrete random variable.
 $\mathbb{Q}, \mathbb{N}, \mathbb{Z}$ countable

Prob. Mass function

Let X be a random variable with range $R_X = \{x_1, x_2, \dots\}$ (finite, countably infinite)

The function

$$P_X(x_k) = P(X=x_k) \text{ for } k=1, 2, \dots$$

is called Prob. Mass function.

Ex: Toss a coin twice, X is number of head.

- Range of $X \rightarrow R_X = \{0, 1, 2\}$

- P_X

$$P_X(0) = \frac{1}{4} \quad P_X(1) = \frac{2}{4} \quad P_X(2) = \frac{1}{4}$$

H T 1

T H 1

T T 0

H H 2

$$P(X) = \begin{cases} \frac{1}{4}, & X=1 \\ \frac{1}{2}, & X=2 \end{cases}$$



$\rightarrow$ probabilities distribution

$\rightarrow$ Cumulative DSN. function

Don't forget

Ex: We have unfair coin $P(H) = p$ and $1-p$ I toss the coin until I get a H.

Y is total number of tosses. What is distribution of Y .

$$R_Y = \{1, 2, \dots, \infty\}$$

$$P_Y(1) = p$$

$$P_Y(2) = (1-p)p$$

$$P_Y(3) = (1-p)^2 p$$

$$P_Y(k) = p \cdot (1-p)^{k-1}$$

$$P_Y(k) = \begin{cases} (1-p)^{k-1} \cdot p & \text{for } k=1, 2, \dots \\ 0 & \text{otherwise} \end{cases} \rightarrow \text{PMF}$$

$$p, p \cdot (1-p), p \cdot (1-p)^2$$

PMF is a prob. measure

1- $0 \leq P_X(x) \leq 1$ for all x

2- $\sum_{x \in R_X} P_X(x) = 1$

3- For any set $A \subset R_X$

$$P(X \in A) = \sum_{x \in A} P_X(x)$$

Ex: Let X be discrete random variable

$$R_X = \{1/2, 3/4\} \quad P_X(1) = P_X(2) = \frac{1}{2} \text{ and } P_X(4) = \frac{1}{2}$$

$$a) P_X(3) = ? \quad \sum_{x \in R_X} P_X(x) = 1 \rightarrow \frac{1}{2} + \frac{1}{2} + P_X(3) = 1 \rightarrow P_X(3) = \frac{1}{2}$$

$$b) P_X(2 \leq x \leq 4) = ? \quad P(X \in A) \text{ and } A \subset R_X \quad P(2) + P(3) = \frac{1}{2} + \frac{1}{2} = 1$$

Independent Random Variables

$$P(A \cap B) = P(A) \cdot P(B)$$

$\rightarrow$ Consider two discrete random variables X, Y . X and Y are independent

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

$$P(Y \in A, X \in B) = P(Y \in A) \cdot P(X \in B)$$

Ex: We roll a die twice and get two numbers X and Y .

a) Find R_X, R_Y and PMF of X and Y .

$$R_X = R_Y = \{1, 2, 3, 4, 5, 6\}$$

$$P_X(1) = \frac{1}{6} = P_X(2) = \dots = P_X(6) \rightarrow P_X(R_X) = 1$$

$$P_Y(1) = \frac{1}{6} \quad P_Y(R_Y) = 1$$

b) Find $P(X=2 \text{ and } Y=6) \rightarrow P(2 \in R_X, 6 \in R_Y) = P_X(2) \cdot P_Y(6) = \frac{1}{6} \cdot \frac{1}{6}$

c) Find $P(X \geq 3 \text{ and } Y \geq 2) \rightarrow (\frac{1}{6} + \frac{1}{6} + \frac{1}{6}) \cdot \frac{1}{6} = \frac{1}{12} \rightarrow P_X(X \geq 3) \cdot P_Y(Y \geq 2)$

d) Let $Z = X+Y$ Range of $Z = ?$

$$R_Z = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$P_Z(2) = P(X+Y=2) \rightarrow P_X(1) \cdot P_Y(1) = \frac{1}{36}$$

$$P_Z(4) = P(X+Y=4) \rightarrow P_X(1) \cdot P_Y(3) + P_X(2) \cdot P_Y(2) = \frac{1}{36} + \frac{1}{36} = \frac{2}{36}$$

Ex: I toss a coin twice

X = number of head I observe

Y = number of heads I observe

$$P(X \leq 2 \text{ and } Y \geq 1)$$

$$\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8} \rightarrow P_X(X \leq 2) \cdot P_Y(Y \geq 1)$$

Independent of n random variables

Consider n random variables $X_1, X_2, \dots, X_n$ we say that $X_1, X_2, \dots, X_n$ are independent

$$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n) = P(X_1=x_1) \cdot P(X_2=x_2) \cdot \dots \cdot P(X_n=x_n)$$

Special Distributions

1- Bernoulli Distribution

A "trial" is a random experiment that has two possible outcomes which can be labeled as success or failure

Simplest discrete random variable

$$P_X(x) = \begin{cases} p & \text{for } x=1 \\ 1-p & \text{otherwise} \end{cases}$$

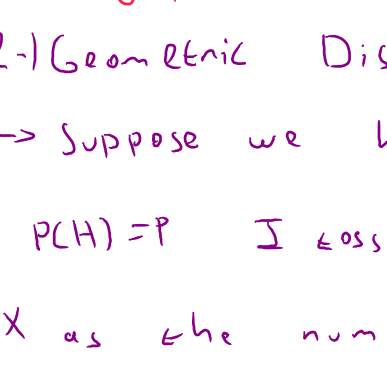
Child $\begin{cases} \text{boy} & 1 \\ \text{girl} & 0 \end{cases}$

Coin $\begin{cases} \text{Tail} & 1 \\ \text{Head} & 0 \end{cases}$

Definition: A random variable X is said to be a Bernoulli Random Variable with parameter p , $X \sim \text{Bernoulli}(p)$ if its PMF

$$P_X(x) = \begin{cases} p & \text{for } x=1 \\ 1-p & \text{for } x=0 \\ 0 & \text{otherwise} \end{cases}$$

$X \sim \text{Bernoulli}(p)$



2- Geometric Distribution

$\rightarrow$ Suppose we have a coin

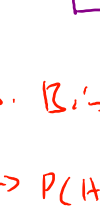
$P(H) = p$ I toss the coin until I observe the first head. We define

X as the number coin tosses. X is said to have geometric distribution.

- Repeating Bernoulli trials until the first success.

$$T \quad T \quad T \quad H \quad \dots \quad k-1 \text{ times} \quad P \cdot (1-p)^{k-1}$$

Definition: A random variable X is said to be a geometric random variable with parameter p , $X \sim \text{Geometric}(p)$ if its PMF is $P_X(k) = \begin{cases} p \cdot (1-p)^{k-1} & \text{for } k=1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$



3. Binomial Distribution

$\rightarrow P(H) = p$ toss a coin n times

X is the number of heads.

$$R_X = \{0, 1, \dots, n\}$$

n independent Bernoulli trials

PMF of X $\binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$

def: A random variable X is said to be binomial random variable with parameters n and p shown as $X \sim \text{Binomial}(n, p)$ and its PMF

$$P_X(k) = \begin{cases} \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k} & \text{for } k=0, 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$$



Binomial is sum of Bernoullies

$X_1, X_2, X_3, \dots, X_n$ are independent Bernoulli(p) random variables

$$X = X_1 + X_2 + \dots + X_n \text{ a Binomial}(n, p) \text{ distribution}$$

Ex: $X \sim \text{Binomial}(n, p)$ and $Y \sim \text{Binomial}(m, p)$

$Z = X+Y$ Find PMF of Z

$$\left. \begin{aligned} X &= X_1 + X_2 + \dots + X_n \\ Y &= Y_1 + Y_2 + \dots + Y_m \end{aligned} \right\} n+m$$

$Z = X+Y = X_1 + X_2 + \dots + X_n + Y_1 + Y_2 + \dots + Y_m$ Since X_i and Y_j are independent Bernoulli trials

$$P_Z(k) = \begin{cases} \binom{n+m}{k} \cdot p^k \cdot (1-p)^{n+m-k} & \text{for } k=0, 1, \dots, n+m \\ 0 & \text{otherwise} \end{cases}$$

4- Pascal (Negative binomial) distribution

- It is the generalization of geometric distribution

- Random experiment of independent trials until observing m successes.

- Coin $P(H) = p$

- Toss the coin until observe m heads

- X as the number of coin tosses.

- X said to have Pascal dist. with parameters (m, p) $X \sim \text{Pascal}(m, p)$

- If $m=1$ Pascal(p) = Geometric dist.

$$R_X = \{m, m+1, m+2, m+3, \dots\} \quad P_X(x)$$

$\rightarrow$ Event $A = \{X=k\}$ $A = B \cap C$ where B event that we have m heads

in first $k-1$ trials, C event that we observe

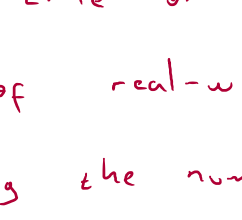
a head in k^{th} trial.

$$P(A) = P(B \cap C) = \underbrace{P(B)}_{\text{Binomial } P} \cdot P(C)$$

$$P(B) = \binom{k-1}{m-1} p^{m-1} (1-p)^{k-m}$$

$$P(C) = \binom{k-1}{m-1} \cdot p^{m-1} (1-p)^{k-m} \cdot p = \binom{k-1}{m-1} p^m (1-p)^{k-m}$$

$X \sim \text{Pascal}(m, p)$



5- Poisson Distribution $\rightarrow$ Zaman analizi var

- Used in cases when we are counting occurrences of certain event in an interval of time or space.

- Approximation of real-world random variable.

- We are counting the number of customers at a certain time period.

- ON average λ customers visit the store

$$P(X=k) = \frac{n!}{k!} p^k (1-p)^{n-k} \quad \text{for } k \in \mathbb{R}_+$$

$$\lim_{n \rightarrow \infty} P(X=k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!} \text{ for } k \in \mathbb{R}_+$$

Poisson distribution

Ex: Number of emails I get model poisson distribution

$\rightarrow$ AVG emails per minute 0.2

$\rightarrow$ What is the prob. that I get no emails in an interval of 5 minutes

λ is Poisson R.V. $\lambda = 0.2, 2 = 1 \quad X = 0$

$$P(X=0) = P_0(0) = \frac{e^{-\lambda} \cdot \lambda^0}{0!} = \frac{e^{-1} \cdot 1}{1} = \frac{1}{e}$$